

Over the years the options market has proven to be a valuable, accurate, and timely indicator of market volatility and changing regimes. Options traders are able to adjust prices quickly based on changing volatility expectations. Analysis can easily infer these expectations through the options prices. This is known as the implied volatility. The question arises then if implied volatility is an accurate and timely estimate of volatility then why cannot analysts just use implied volatility from the options market rather than use results from these models? The answer is simple. Unfortunately, implied volatility estimates do not exist for all stocks. The options market at the stock level is only liquid for the largest stocks. Accurate implied volatility estimates do not exist across all stocks. Fortunately, the options market still provides valuable information that could be extended to the stock level and help provide accurate forward looking estimates, and in a more timely manner than the other historical techniques. This also provides ways for algorithms to quickly adjust to changing expectations in real-time and provide investors with improved trading performance.

The HMA-VIX technique consists of adjusting the stock's trailing volatility by the ratio of the VIX index to the SP500 trailing volatility plus a correction factor. The ratio of the VIX to the SP500 realized shows whether the options market believes that volatility will be increasing or decreasing. However, since the VIX usually trades at a premium of 1.31 to the SP500 trailing volatility we need to include an adjustment factor to correct for this premium. If the VIX Index/SP500 realized volatility > 1.31 then we conclude that the options market believes volatility will be increasing and if the VIX Index/SP500 realized volatility < 1.31 then we conclude that the options market believes volatility will be decreasing.

The increasing/decreasing expectation obtained from the options market is then applied to individual stocks. A comparison of the VIX implied to the SP500 trailing is shown in [Figure 6.4](#).

The advantage of incorporating the implied expectations into our real-time volatility estimator is that if there is a sudden market event that will affect volatility it will almost immediately be reflected in the HMA-VIX measure. The historical models (HMA, EWMA, ARCH and GARCH) will not react to the sudden market event until after this event has affected stock prices. Thus the historical models will always be lagging behind the event to some degree. Furthermore, if the options market is anticipating an event that has not yet occurred, and has priced the uncertainty of the event into its prices, the HMA-VIX model will also reflect the anticipated event and increased uncertainty prior to that event taking